October 8, 2026

Air Commodore Dr. Saifur Rahman Bakaul, BUP, ACSC, psc
Chairman, SPARRSO
Agargaon, Dhaka-1207

Subject: Application for SPARRSO Research Fellowship: In-House ADCS (Attitude Determination and Control System) Board for Satellites/CubeSat

Dear Sir,

I am Md Tasbeer Ahmed, a Lab Instructor and MSc student in Electrical and Electronic Engineering at North South University, and a Student Researcher at the NIRO Lab. I respectfully apply for the Junior Research Fellowship under SPARRSO's priority areas of satellite subsystems (ADCS), control theory and sensor fusion, and microcontroller and PCB design.

I am one of the project leaders of the ADCS team of the **NSU-AI-SAT** project, Bangladesh's entry in the **APSCO CubeSat Competition (ACC)**, for which SPARRSO coordinated the national selection. I also work on link budget analysis and antenna design for its communication subsystem. My undergraduate thesis on the uplink/downlink budget for HD video transmission with DVB-S2 was first runner-up at the Undergraduate Thesis Paper Contest, **IEEE Student Led Con 2023**.

At SPARRSO, I aim to design, build and test an in-house, STM32-based ADCS board, together with the attitude estimation and control algorithms that run on it, so that Bangladesh has locally designed and adaptable satellite hardware.

I believe this work could help Bangladesh in the following ways:

- **Less dependence on imports:** A locally designed ADCS would reduce the need to buy costly foreign integrated units.
- **Local capability:** It would build skills in attitude control, PCB design and embedded flight software among Bangladeshi engineers and students.
- **A reusable base:** Future CubeSat missions and universities could inspect, modify and reuse the design and software.

I would be grateful for the opportunity to join as a SPARRSO Research Fellow and carry out this work under the guidance of SPARRSO's experts, and I am committed to contributing fully to the fellowship's goals. Thank you for your kind consideration.

Sincerely,

Md Tasbeer Ahmed
Lab Instructor and MSc Student | Dept. of ECE, North South University
tasbeer.ahmed.201@gmail.com | +880 1775-452647